LPCOpen v2.xx Keil uVision4 quickstart guide for all platforms

Keil Quick start introduction

This quick start guide will guide you through the steps of using the Keil MDK-ARM toolchain with the LPCOpen projects for your desired device and platform.

Download and install the Keil LPCOpen package for your platform

If you haven't downloaded an LPCOpen package yet, go the LPCOpen home page on www.nxp.com to download the package for the device/platform you want to use. You will need to download a package that supports the Keil toolchain with this quick start guide.

Once you have downloaded your package, you will need to create a work directory somewhere and unzip the package in that directory.

We highly recommend using /nxp/lpcopen as the install area, although you can install it anywhere.

Windows users only: Install LPCOpen close to the root drive on your system as path names can exceed the character limit if installed under too many directories. You'll get project errors if this happens.

Needed Keil tools

To use the Keil projects in LPCopen, you will need the Keil MDK-ARM toolchainand a ULINK/ME debugger. You can get an emulation version of the Keil MDK-ARM here. All LPCOpen projects for Keil are preconfigured for the ULINK/ME debugger. You can get more information on that debugger here.

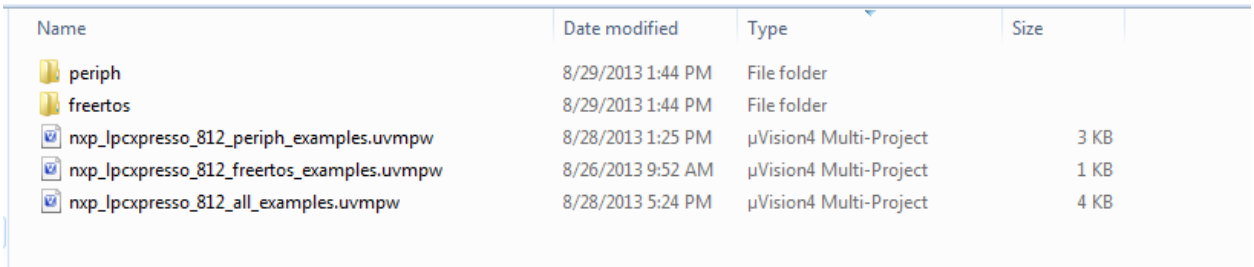
Keil project startup

To bring up the LPCopen v2.xx projects using Keil uVision4, you will need to navigate to the Keil project area for the board you are using. The project file areas for each chip family are listed in the table below:

Device/Platform	LPCOpen project directory
LPC8xx family of devices	applications\lpc8xx\iar_ewarm_projects
All LPC11xx families of devices	applications\lpc11xx\iar_ewarm_projects
All LPC13xx families of devices	applications\lpc13xx\iar_ewarm_projects
LPC175x/6x, LPC177x/8x, and LPC40xx families of devices	applications\lpc17xx_40xx\iar_ewarm_projects
LPC18xx, and LPC43xx families of devices	applications\lpc18xx_43xx\iar_ewarm_projects

Depending on which LPCOpen package you download, you'll have one or more products in the project file area. Go into the directory for the product you want to use.

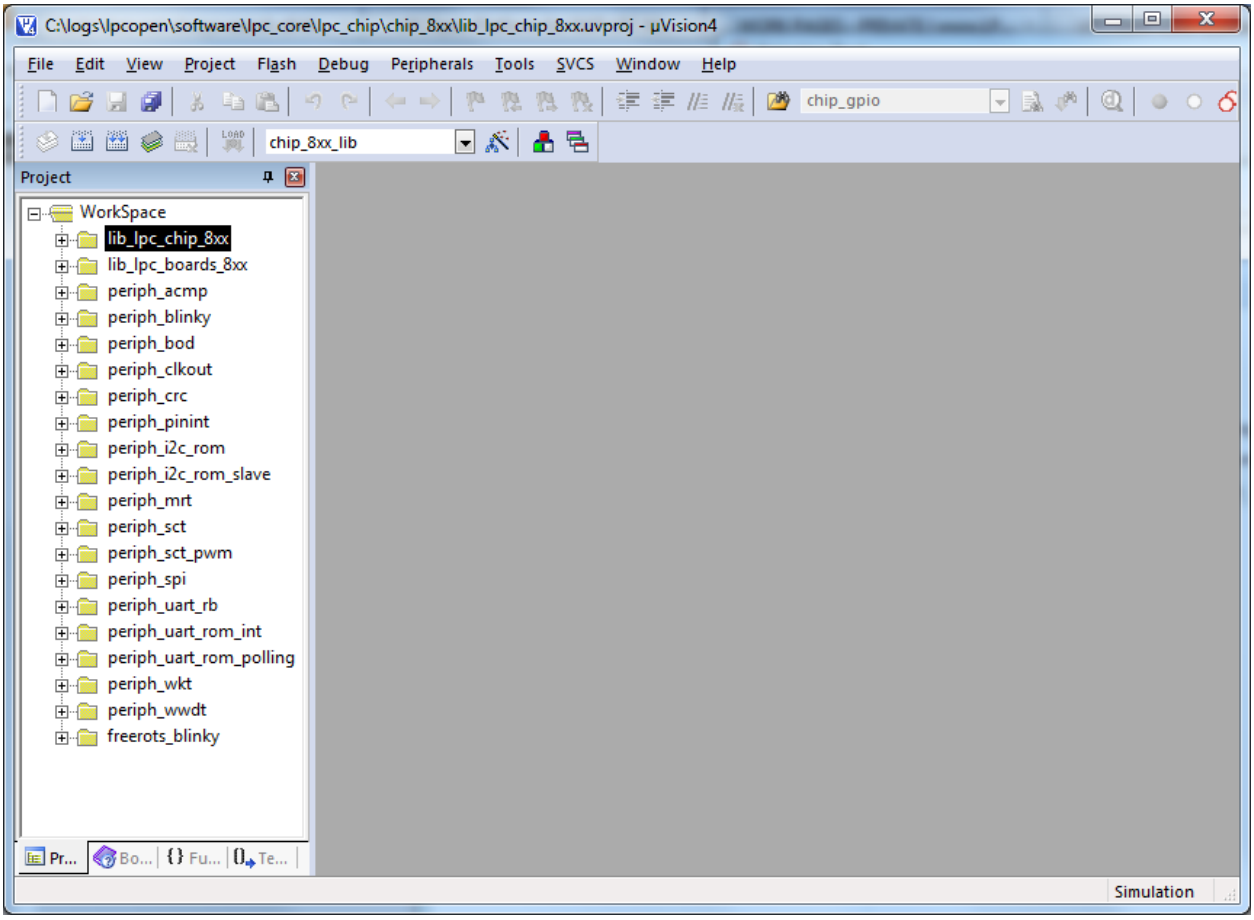
For example, if you want to use the LPCXpresso LPC812 board with Keil uVision4 and have download the LCP812 Keil package, enter the 'applications\lpc8xx\keil_uvision_projects\nxp_lpcxpresso_812' directory to enter the LPCXpresso LPC812 Keil project area. After entering the project area, you'll see a collection of multi-project files similar to the list below. (This list may vary per platform)



Name	Date modified	Type	Size
periph	8/29/2013 1:44 PM	File folder	
freertos	8/29/2013 1:44 PM	File folder	
nxp_lpcxpresso_812_periph_examples.uvmpw	8/28/2013 1:25 PM	µVision4 Multi-Project	3 KB
nxp_lpcxpresso_812_freertos_examples.uvmpw	8/26/2013 9:52 AM	µVision4 Multi-Project	1 KB
nxp_lpcxpresso_812_all_examples.uvmpw	8/28/2013 5:24 PM	µVision4 Multi-Project	4 KB

Simply double click the <platform>_all_examples.uvmpw project file to bring up Keil with the LPCopen projects. In the image above, the file to clock would be 'nxp_lpcxpresso_812_all_examples.uvmpw'.

After Keil finishes loading, you'll have a workspace with all projects for that platform ready for build and deployment.



Building the libraries

Start by selecting the lib_lpc_chip_<device> project and then building it. To select the project, right click on the project in the list and select 'Set as active project'. To build the project after it is selected, select 'Build' from the Project menu. This will build the chip library needed by all other projects.

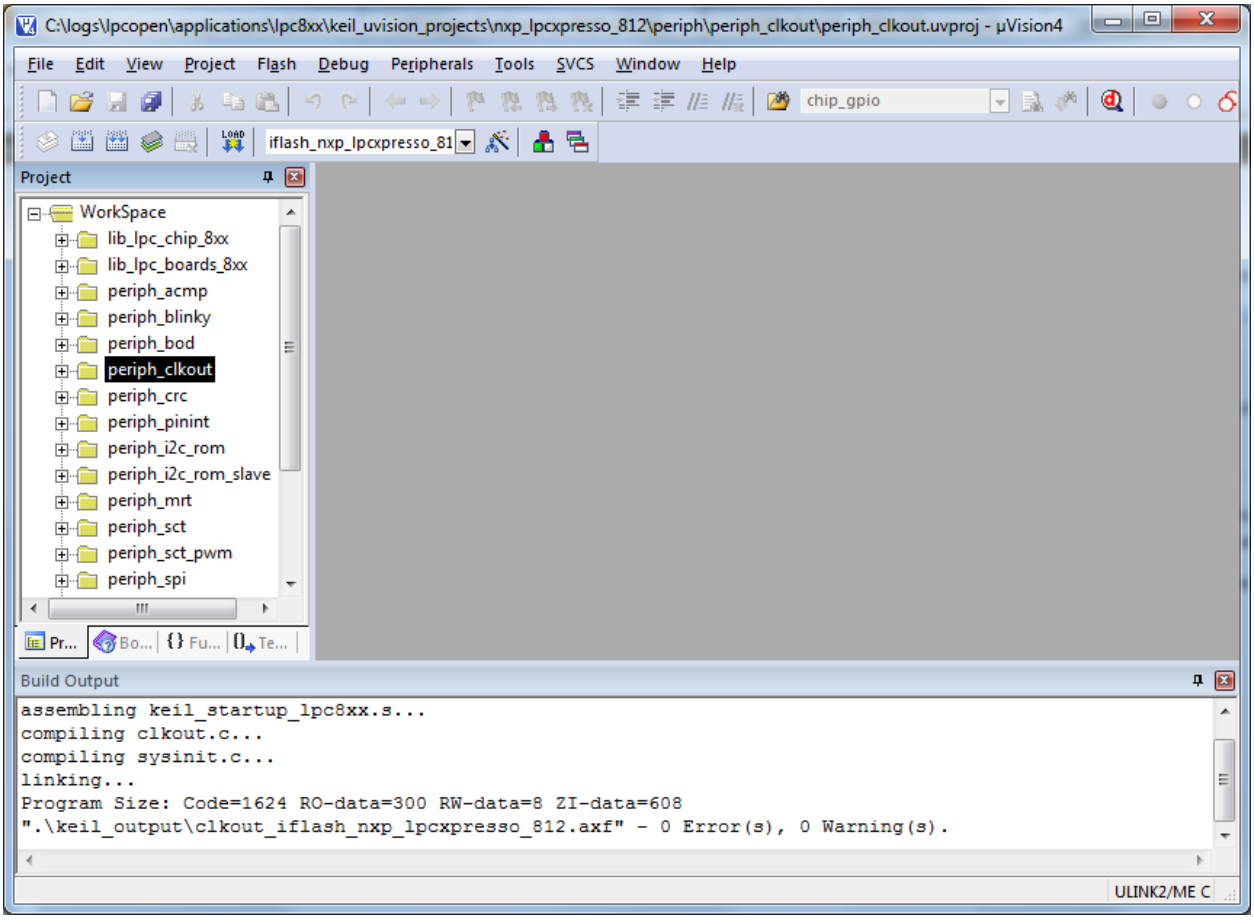
After the chip library is built, select the lib_lpc_boards_<platform> project and then build it. This will build the board library needed by other projects.

Note that some projects may require additional libraries - for example, LPCUSBLUB also requires the USB librarires to be build prior to using the USB examples. These libraries will appear in the projects lists for platforms that suppor them, so build them as needed. If you forget to build the necessary library, you will get a linker error in your application with an error indicating a library is missing.

The chip and board library are used by most other projects as libraries and normally only need to be built once. If you changes chip or board code, you must manually select and rebuild these libraries prior to building your application.

Building an example

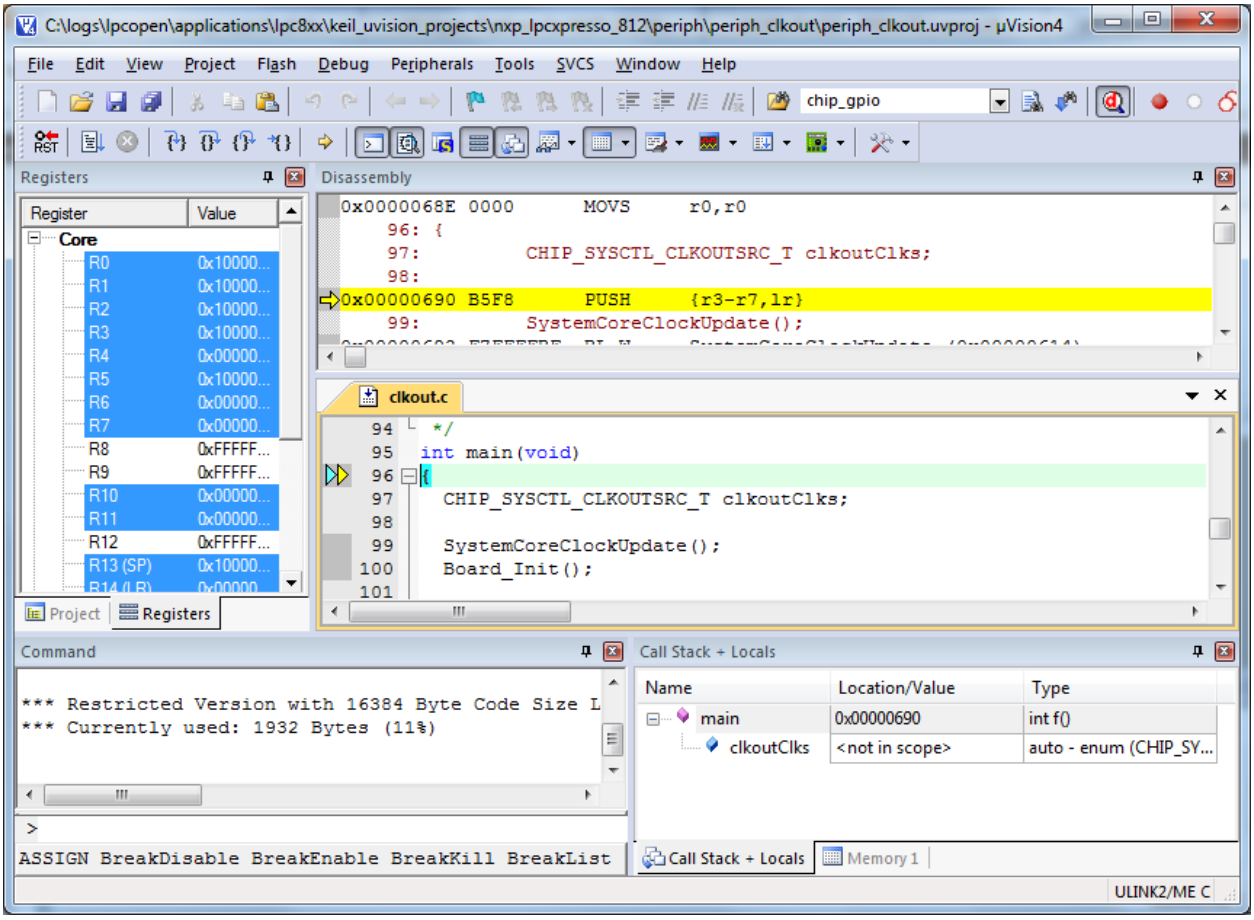
To build an example, select the example as the active project and then select 'Build' from the project menu. The project will build and is ready for debug.



Starting a debug session

Power up your board that you are using the LPCOpen and Keil uVision4. Connect the ULINK/ME debugger to the debug (SWI/JTAG) connector on the board. Connect a USB cable between the ULINK/ME and the host machine running Keil uVision4. Install drivers if needed for the ULINK/ME debugger.

From the Debug menu, select 'Start/Stop Debug Session' to program the image to the board and start it. For most projects, the debugger will automatically either stop at main() or at the startup code. Select commands from the Debug menu to run the program, set breakpoints, stop debugging, etc.



That’s all there is to it!

How to Reach Us

Home Page: www.nxp.com
Web Support: www.nxp.com/support

USA/Europe or Locations Not Listed:
NXP Semiconductor
Technical Information Center, EL516
2100 East Elliot Road
Tempe, Arizona 85284
+1-800-521-6274 or +1-480-768-2130
www.nxp.com/support

Europe, Middle East, and Africa:
NXPHalbleiter Deutschland GmbH
Technical Information Center
Schatzbogen 7
81829 Muenchen, Germany
+44 1296 380 456 (English)
+46 8 52200080 (English)
+49 89 92103 559 (German)
+33 1 69 35 48 48 (French)
www.nxp.com/support

Japan:
NXP Semiconductor
ARCO Tower 15F
1-8-1, Shimo-Meguro, Meguro-ku,
Tokyo 153-0064, Japan
0120 191014 or +81 3 5437 9125
support.japan@nxp.com

Asia/Pacific:
NXP Semiconductor Hong Kong Ltd.
Technical Information Center
2 Dai King Street
Tai Po Industrial Estate
Tai Po, N.T., Hong Kong
+800 2666 8080
support.asia@nxp.com

NXP and the NXP logo are trademarks of NXP B.V. All other product or service names are the property of their respective owners. All rights reserved. © 2016 NXP B.V.

Document Number:
XXXXXXXXXX